

A Comparison of Test Statistics for the Equality of Covariance Matrices in High-Dimensional Data and an Application to Microarray Data

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ABSTRACT

This study compares the performance of five statistical tests T_1 by Schott (2007), T_2 by Srivastava and Yanagihara (2010), T_3 by Ahmad (2017), T_4 by Qayed and Han (2020), and T_5 by Sun et al. (2022) for testing the equality of covariance matrices in high-dimensional multivariate normal data. Monte Carlo simulations were conducted under five matrix structures: (AR1), Compound Symmetry, Simple, Banded Toeplitz, and Variance Components. Performance was evaluated based on type I error control and power. The results indicate that T_3, T_4 and T_5 effectively controlled the type I error rate, whereas the others did not. The test statistic T_3 the most reliable results, as it exhibited the highest power in several scenarios simulation study.

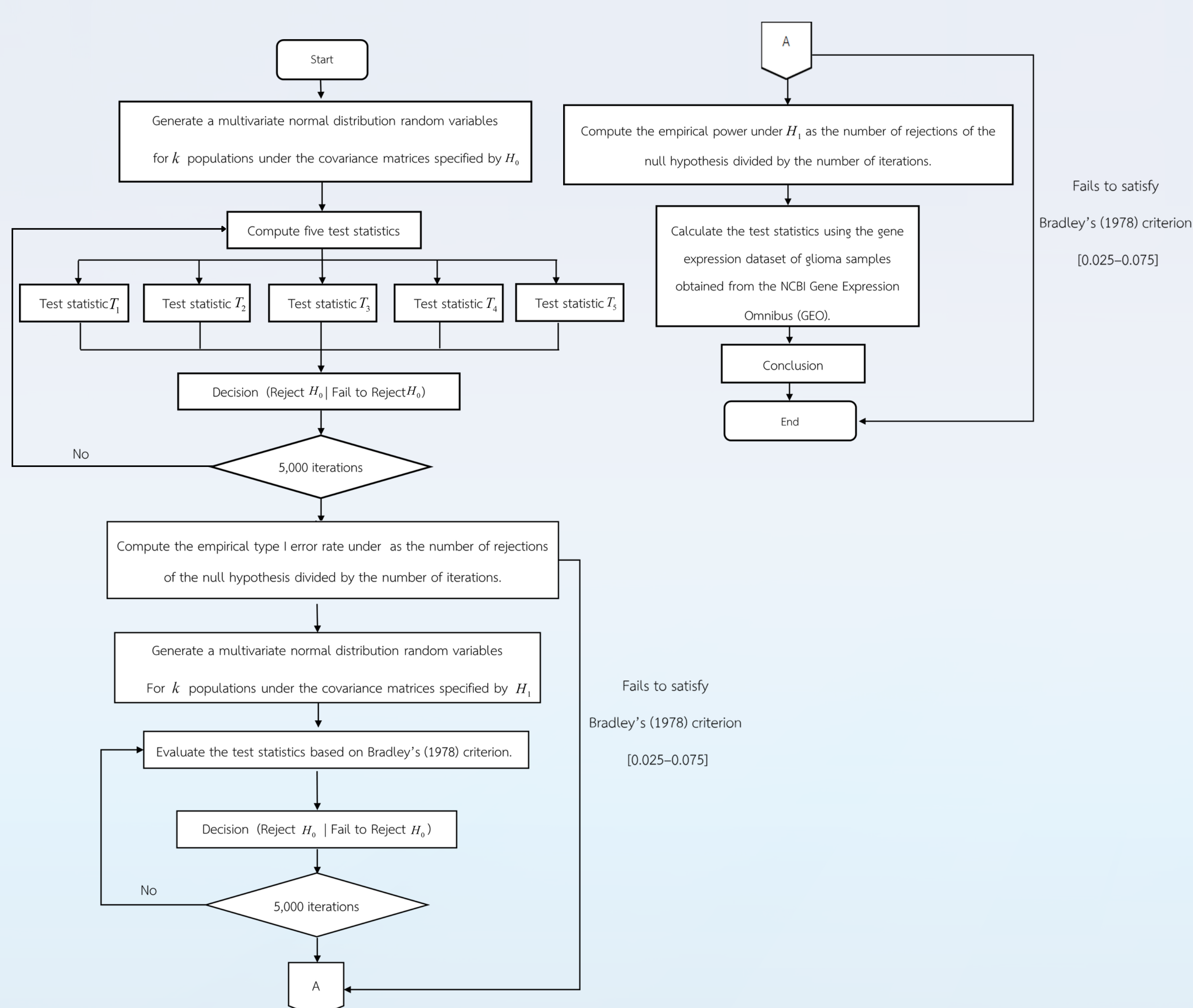
INTRODUCTION

Classical tests for equality of covariance matrices, such as Box's Test and the Likelihood Ratio Test, require $p < n$ and are not applicable in high-dimensional settings where $p \geq n$. To overcome this limitation, several high-dimensional test statistics have been proposed to control the type I error while maintaining empirical power. This study compares five such tests under various covariance structures, evaluating their type I error rate control and empirical power to provide guidance for practical high-dimensional data analysis.

OBJECTIVE

This study focuses on statistical tests for assessing the equality of covariance matrices across multiple populations in high-dimensional data. The five test statistics considered are those proposed by Schott (2007), Srivastava and Yanagihara (2010), Ahmad (2017), Qayed and Han (2020), and Sun et al. (2022). The study also compares the performance of these five tests based on their ability to control the type I error rate and their power.

METHODOLOGY



RESULTS AND DISCUSSION

The results illustrate the empirical type I error rate and empirical power for the First-Order Autoregressive (AR1) structure, where $k = 3$ across five different covariance matrices structures Figure. 1 are presented. It was found that T_4 performed best in controlling the empirical type I error rate within Bradley's criterion, yielding results consistent with other covariance matrices structures. Regarding the empirical power, T_3 exhibited the highest estimated power. Notably T_1 and T_2 failed to control the empirical type I error rate; therefore, their empirical power was not further evaluated. These findings are consistent with those observed for the Banded Toeplitz and Variance Components structures.

The simulation results for the case of $k = 4$ are consistent with those observed when $k = 3$.

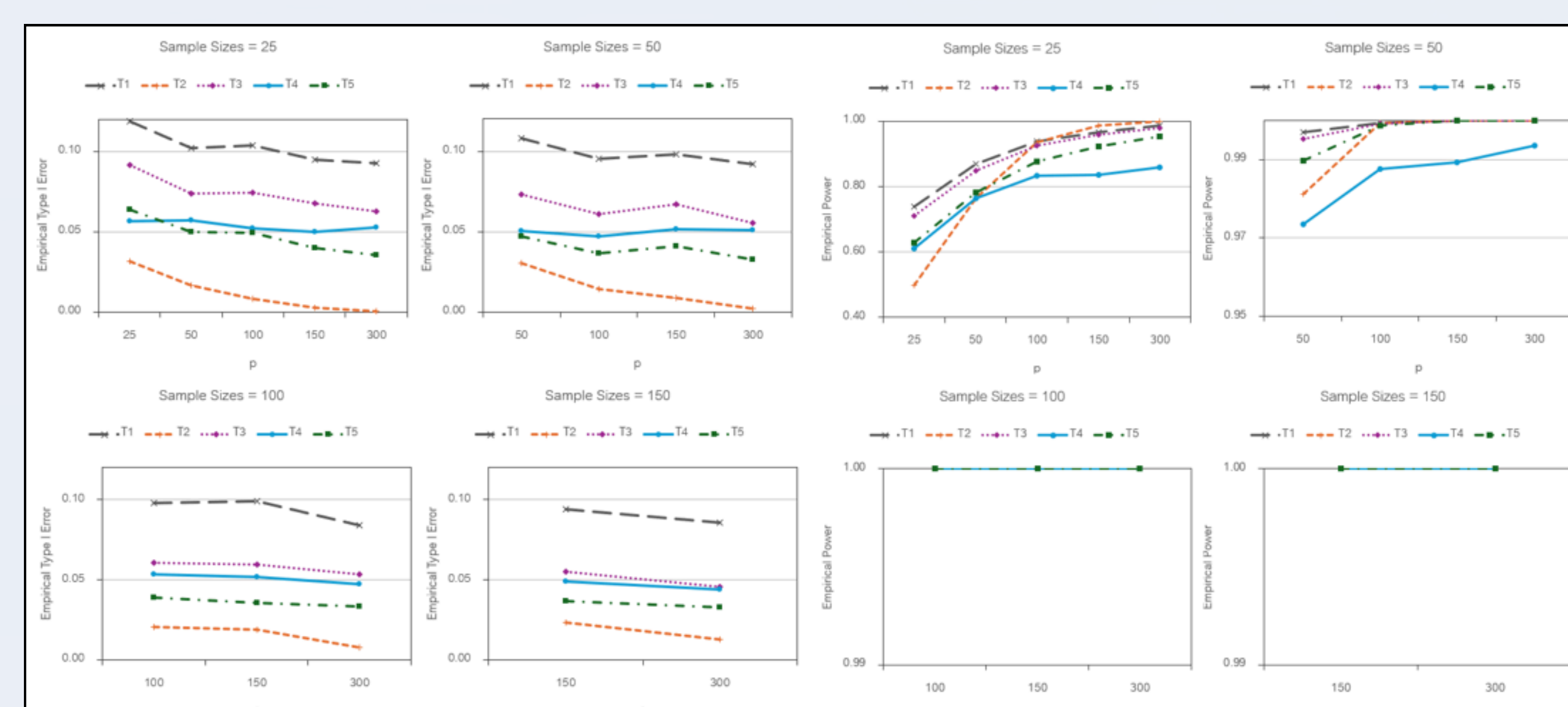


Figure. 1 Empirical Type I Error under H_0 and Empirical Power under H_1 in First-order Autoregressive (AR1) structure when $k=3$

In addition, the proposed methods were applied to a glioma gene expression dataset. Variables were selected based on their coefficients of variation restricted to the interval $[0.25-0.5]$. To evaluate testing performance on real data, the dataset consists of $k = 4$ groups with $p = 24785$ variables and a total sample size of $n = 180$. The group sample sizes are $n_1 = 26$, $n_2 = 81$, $n_3 = 50$ and $n_4 = 23$, and the results reported in Table 1 indicates that T_3 and T_5 reject H_0 (p-value < 0.05), indicating that the covariance matrices are not equal across groups at least two groups, whereas T_4 fails to reject (p-value > 0.05) suggesting that the covariance matrices can be considered equal.

	T_3	T_4	T_5
Test Stat	2.0469	5.9589	29.3143
p-value	0.0203	0.1136	< 0.001

Table 1 Test statistics and p-values of the statistical tests T_3, T_4 and T_5 in the gene expression dataset of glioma brain tumors

CONCLUSION

The results indicate that T_3, T_4 and T_5 adequately control the type I error rate, whereas T_1 and T_2 perform poorly in high-dimensional settings. Among the valid methods, T_4 is closest to the significance level of 0.05.

In terms of statistical power, T_3 achieved the highest power under several covariance structures, including First-Order Autoregressive (AR1), Simple, and Variance Components, followed by T_5 and T_4 , respectively. Under Compound Symmetry and Banded Toeplitz structures, T_4 exhibited the highest power, followed by T_3 and T_5 .

Application to real gene expression data indicates the covariance matrices are not equal across groups at least two groups.

In summary, T_3 is the most efficient and reliable test statistic due to its ability to control the type I error rate and maintain high power in high-dimensional data analysis.

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